United International University (UIU)

Dept. of Computer Science & Engineering (CSE)

Mid Exam: Spring 2026
Course Code: CSE 2215, Course Title: Data Structures and Algorithms 1
Total Marks: 30 Duration: 1 hour 30 minutes

Answer all questions. Marks are indicated in the right side of each question.

[Any examinee found adopting unfair means will be expelled from the trimester/program as per UIU disciplinary rules.]

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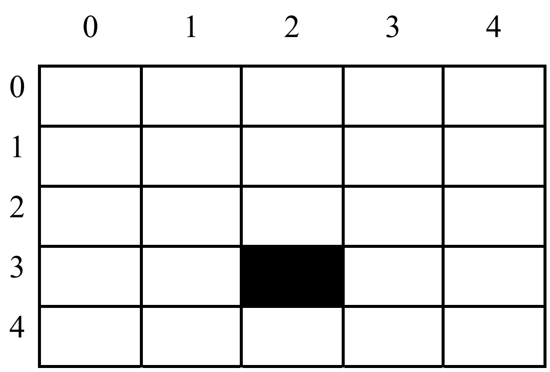
1. a. Which sorting algorithm is more suitable for a **nearly sorted array**: **Insertion Sort** or **Selection Sort**? Justify your answer with appropriate reasoning based on time complexity and behavior. **[3]**

   b. Given the array: **[3]**
   A = [5, 1, 4, 1, 3, 2, 5, 0]
   i. Apply **Counting Sort** to sort the array.
   ii. Show the **frequency (count) array**.
   iii. Show the **final sorted output array**.

2. a. The address of the marked cell is **552**. If the base address is **500** and the component length is **4**, does this array follow **row-major order** or **column-major order** memory addressing? **[3]**



   b. Suppose you are given the following datasets. For each one, decide whether **Binary Search is applicable or not**. If not, suggest a better approach. **[3]**
   i. [45, 12, 78, 34, 23, 56]
   ii. ["zebra", "yak", "xray", "wolf", "vulture"]
   iii. A list of names entered randomly by users in real-time

3. a. Determine whether the following statements are true or false. Justify your answer. **[3]**
   i. **$n^2 + 3n + 1$ is O(n)**
   ii. **log(n)** grows faster than **n**
   iii. **O(1)** means the algorithm takes exactly one step

b. Determine the asymptotic time complexity of the following code segment and explain why the inner loop does not change the time complexity significantly. **[3]**

```
for(int i = 1; i <= n; i++) {
   for(int j = 1; j <= 5; j++) {
      printf("*");
   }
}
```

4. Write a recursive function, "**oddArray**" that will print the **odd elements** from an **integer array** of size **n**. **[4]**
   For example, if A[5] = {1, 3, 4, 6, 5}, calling the function, **oddArray** will print **1, 3, 5**.

5. a. Is **random access** possible in a linked list? Why or why not? **[1]**

   b. What will be the **output** generated by the following code segment for the given **single linked list**? Consider the first node as head. Explain your answer. **[3]**

   24 -> 45 -> 54 -> 6 -> 22 -> 11 -> 8 -> 12

```
struct node* temp = head;
while(temp != NULL)
{
   printf("%d", temp->value);
   if(temp->value%2==0)
         temp = temp->next;
   else
         temp = temp->next->next;
}
```

   c. Write a pseudocode **void print_n_from_end(struct node * head, int n)** to find and print the value of the **nth node from the end** of a **double linked list**. **[4]**